

KAVACH AI

India's AI-Powered Digital Safety Shield

Defeating Counterfeiting, Fraud & Digital Arrest Scams

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Track: Smart Cities / Public Safety / Digital Trust / Geospatial Law Enforcement

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1. Problem Statement & Executive Context

The Growing Public Safety Crisis:

India registered over **1.14 million cybercrime complaints** in 2023 — a 60% surge from 2022. Among the most dangerous emerging threats are '**Digital Arrest**' scams, where victims are psychologically held hostage over video calls by criminals impersonating CBI, ED, or Police officers. The Ministry of Home Affairs reported citizens defrauded of over **Rs 1,776 crore** in the first nine months of 2024 alone.

Concurrently, counterfeit currency (Fake Indian Currency Notes — FICN) continues to undermine financial systems, with high-quality Rs 500 fake notes bypassing traditional manual detection. Existing law enforcement frameworks are reactive, stepping in only post-complaint rather than neutralizing threats at the point of contact.

2. Proposed Solution — KAVACH AI

KAVACH AI is an integrated, multi-agent AI public safety platform designed to equip citizens, financial institutions, and law enforcement agencies (LEO) with real-time proactive tools to detect, disrupt, and neutralize digital fraud and counterfeit currency.

- **Citizen Fraud Shield:** Multilingual conversational advisory in 12 regional languages providing real-time call/message threat verdicts & 1-click reporting to NCRB 1930.
- **Digital Arrest Scam Detector:** ML-powered transcript & speech classifier trained on 15 call categories to detect impersonation, fear tactics, and urgency indicators.
- **Counterfeit Currency Agent:** Computer vision forensics analyzing microprint, intaglio edge sharpness, HSV color profiles, and printer frequency artifacts across denominations.
- **Fraud Network Graph AI:** Graph algorithms (Louvain Community Detection, PageRank) mapping money mule clusters, suspect nodes, and organized campaign infrastructure.
- **Geospatial Crime Hotspots:** Spatial analytics (DBSCAN & Kernel Density Estimation) identifying cybercrime hotspots and guiding police patrol prioritization.
- **LEO Command & Court Evidence:** End-to-end case management generating court-admissible PDF intelligence packages.

3. Hero Feature Showcase — Distinguishing the 3 Dashboards

KAVACH AI provides **three completely distinct, role-tailored hero dashboards** designed for Citizens, Law Enforcement Officers (LEO), and System Administrators. Each dashboard serves a unique operational purpose:

Dashboard Role	Target Audience & Operational Goal	Distinguishing Hero Features & Capabilities
CITIZEN DASHBOARD Public Safety Shield	Citizens, families, and banking customers verifying suspicious calls, messages, or notes.	• Scam Detector: Text & Call Transcript classifier. • Currency Verification: 6-layer CV note checks. • Media Forensics: Deepfake face & voice cloning. • RAG Advisory: 12 Regional Indian languages. • 1-Click NCRB 1930: Direct official cybercrime reporting.
LEO DASHBOARD Command & Investigation	Law Enforcement Officers, Cyber Cells, and Financial Investigators.	• Case Management: Lifecycle status & suspect vault. • Fraud Network Graph: Louvain mule ring clustering. • Geospatial Hotspots: DBSCAN & KDE density maps. • Court-Admissible Evidence: Export PDF packages.
ADMIN DASHBOARD Platform Governance	System Administrators, MHA Authorities, & Network Operators.	• User & Role Control: Elevate roles (Citizen→LEO). • Emergency Broadcast: Dispatch priority SMS/Alerts. • System Audit Log: Track all API & security events. • Health Diagnostics: Real-time FastAPI & DB status.

Role Isolation & Security Controls:

Each dashboard is strictly isolated via JWT role-based access control (RBAC) middleware (`LEO-only`, `ADMIN-only`). Citizens have no visibility into police case files or system audit registries, ensuring data confidentiality and public safety integrity.

4. Live Dashboard UI Screenshots Tour

Actual screenshots captured directly from each distinct role view in the live application:

1. Citizen Safety Shield & Scan Portal View:



2. Law Enforcement Officer (LEO) Command View:

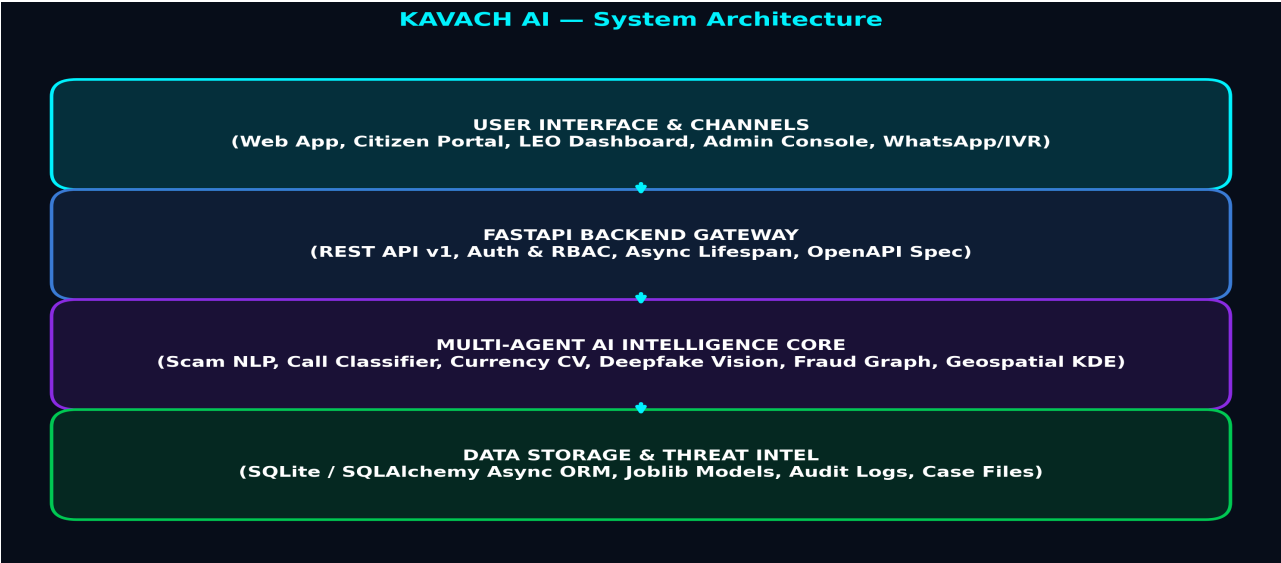


3. Admin Platform Governance & User Control View:

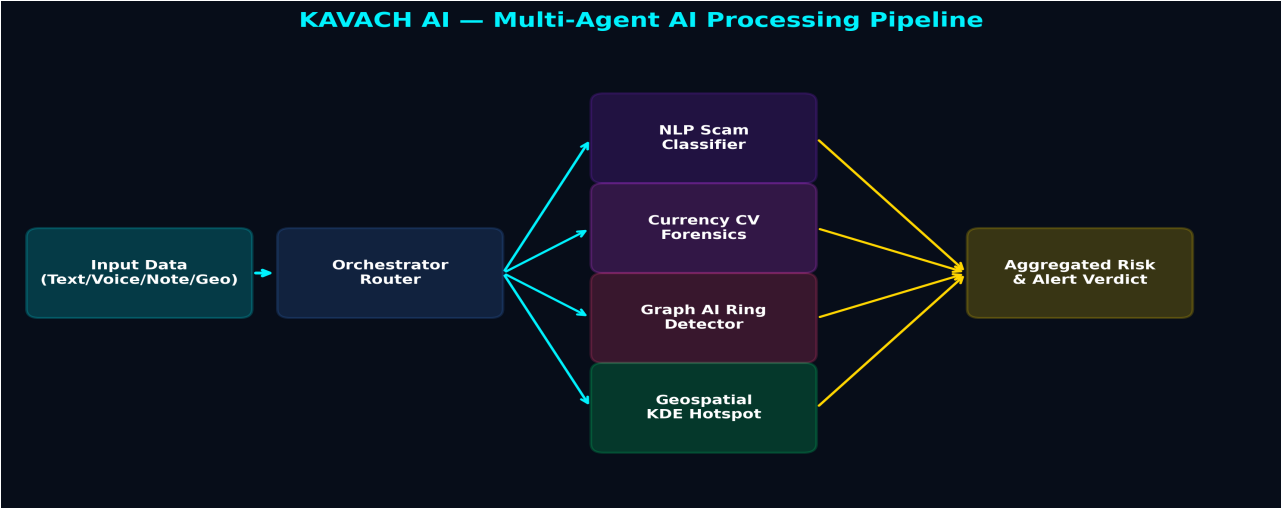


5. System Architecture & Processing Workflows

System Architecture Diagram:



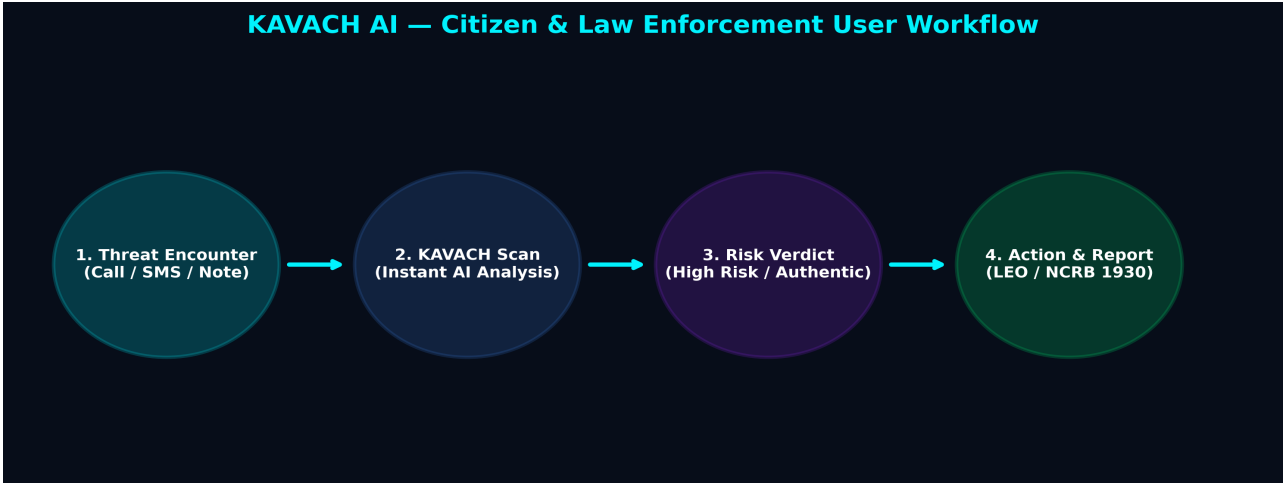
Multi-Agent AI Processing Pipeline:



6. Technology Stack & Technical Matrix

Component	Technologies Used	Key Purpose
Frontend	HTML5, Vanilla JS, TailwindCSS, Iconify, GSAP	Responsive Citizen, LEO & Admin Dashboards
Backend API	FastAPI, Uvicorn, Python 3.11, Pydantic v2	Async REST APIs & Microservices Layer
AI & ML Models	Scikit-Learn, NetworkX, SciPy, NumPy, Joblib, Whisper	Scam NLP, Call ML, Graph AI, Geospatial
Computer Vision	OpenCV, Pillow (PIL), FFT Frequency Analysis	Currency Note Forensics & Deepfake Detection
Database & Auth	SQLite, SQLAlchemy Async ORM, PyJWT, Bcrypt	Persistent Data & Role-Based Access

User Processing Journey Workflow:



7. Future Scope & Strategic Roadmap

- **WebRTC Live Call Interception:** Integrating real-time audio stream monitoring directly on mobile devices to warn citizens mid-call during active digital arrest attempts.
- **Direct Carrier & WhatsApp Cloud API Integration:** Webhook bridges connecting telecom service providers and Meta WhatsApp Business API for instant threat blocking.
- **Distributed Graph & Database Scaling:** Upgrading SQLite persistence to enterprise PostgreSQL + Neo4j for multi-billion transaction graph analysis.
- **Hardware UV Sensor Integration:** Calibrating computer vision models for specialized UV/IR camera hardware on banking note counting machines.

8. Socio-Economic Impact & Evaluation Alignment

KAVACH AI addresses the core evaluation criteria of the ET AI Hackathon 2026:

- **Innovation (25%):** Multi-agent fusion uniting NLP, Computer Vision, Graph AI, and Geospatial analytics into a single public safety shield.
- **Business & Social Impact (25%):** Safeguarding citizen life savings against psychological digital arrests and preventing counterfeit currency proliferation.
- **Technical Excellence (20%):** Self-contained, zero-external-dependency production architecture with 25+ REST endpoints and trained ML models.
- **Scalability & UX (30%):** Async FastAPI backend, 12 regional languages, and intuitive dark-mode glassmorphism interface.

Conclusion:

Project **KAVACH AI** by Team **BitBuddies** provides a production-grade, multi-agent AI safety shield that shifts public safety from reactive post-crime reports to real-time predictive threat neutralization for citizens, law enforcement, and financial institutions.